



AI-assisted Software Development

Trends and Lessons

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GitHub Next

Research, Build, Ship, Learn

GitHub Next

GitHub Next investigates the future of software development.

We are a team of researchers and engineers at GitHub, exploring things beyond the adjacent possible. We prototype tools and technologies that will change our craft.

We identify new approaches to building healthy, productive software engineering teams.



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Students Big Questions!

"What will coding with AI look like in the future?"

"Does AI mean that code will be obsolete soon?"

"How should I use AI tools in software development?"



We live in incredible, disturbing times

- **AI, indexing and traditional dev tools** make us all magicians when it comes to code
- But the magic power comes with many difficulties!



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- But the magic power comes with many difficulties!



Some Magic Wands



Meet Cody Your
AI-Assistant



CURSOR



GitHub
Copilot



AI code completions

JS test.js 1 ●

JS test.js >  calculateDaysBetweenDates

```
1  function calculateDaysBetweenDates(begin, end) {  
    var beginDate = new Date(begin);  
    var endDate = new Date(end);  
    var days = Math.round((endDate - beginDate) / (1000 * 60 * 60 * 24));  
    return days;  
  }  
2
```



sample code completions



llm command line



AI Dev Tools are changing fast

- 2022: **GitHub Copilot Completions** launched. Today:
 - 1.8M+ paying users
 - 50%+ of users' code is written by Copilot
- 2023: More AI developer tools launched
 - Summarization, advice, code review
 - Cody and others
- 2024: **GitHub Copilot Workspace, Cursor Composer, Devin** launched
 - Shift to task-orientation
 - Cursor Composer brought task-orientation to the IDE
- 2025: Cursor/VS Code **Agent-Mode, Claude Code** and others launched
 - Increases the available automation
- 2025: **Automated app development** starts to take off
 - GitHub Spark, Bolt, Gemini, Lovable, Cogna, v0, ...



Where does GitHub Next work?

AI APPLICATIONS & AI DEV TOOLS

AI DEV SERVICES (e.g. indexing, summaries)

FINE TUNING & INFERENCE & GPUS & HOSTING &

MODELS



The Big Trends I'm seeing in AI Dev Tooling

- AI-assisted Pinpoint Surgery
- AI-assisted Task Oriented Surgery
- Increasing AI Automation
- AI-Assisted App Development



Theme 1: Pinpoint Surgery

Original GitHub Copilot completes code in your editor

- You are already located at a pinpoint of change
- You accept/reject small bits
- You have to type quite a lot
- Copilot sort of infers what you are trying to do

This is magical: a finely honed surgical assistant



Other Pinpoint Tools

- Type-checking
- Autocomplete
- On-the-fly checking
- Code navigation
- Code search
- Security checking



Other Pinpoint Tools

- Type-checking
- Autocomplete
- On-the-fly checking
- Code navigation
- Code search
- Security checking
- **AI Code completion**
- **AI Code summarization**
- **AI code chat**
- **AI Code review**
- **AI Code-fix suggestions**
- **AI Test generation**



Theme 2: From Pinpoint Surgery to Tasks

Software activity begins with words and intent

Before a code change, before a pull request, there are words - a discussion, a bug report, a feature request, an engineering task, a meeting, a design document, an idea.



Theme 2: From Pinpoint Surgery to Tasks



Theme 2: From Pinpoint Surgery to Tasks



Theme 2: From Pinpoint Surgery to Tasks



Theme 2: From Pinpoint Surgery to Tasks



Theme 2: From Pinpoint Surgery to Tasks

Changing perspective

Pinpoint Surgery → Whole Operation

Single Point of Change → Whole Repository

Implicit Task → Explicit Task



Theme 2: From Pinpoint Surgery to Tasks

"Add a refresh button to the main page"

"Create end-to-end tests using Playwright"

"Set up continuous integration using GitHub Actions"

"Set up production resources in Azure using Terraform"

"Read coverage report and add unit tests to repo"

"Create a website for UK house prices"



Demo - Task orientation



Theme 3: Automation with SWE-Agents

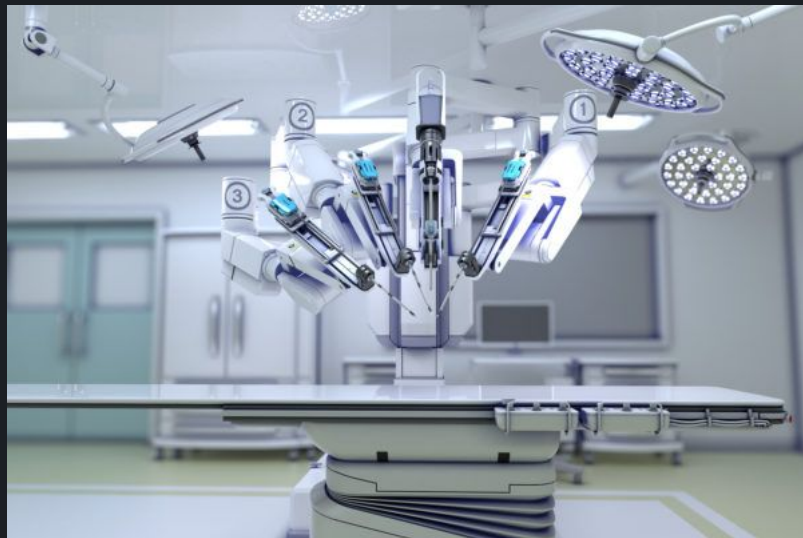
SWE-Agents are AI agents that have multiple tools to operate on your code.

Can be unreliable, the “sweet zone” requires skill to find.

But improving all the time and competent at some things.

Learn and use them!

Don't let them operate on you!



Theme 4: AI-Assisted App Development

"Create me a multiplayer online TicTacToe game"

The Gemini logo features the word "Gemini" in a blue-to-purple gradient font, with a small purple four-pointed star above the 'i'.The Cognia logo consists of two blue trapezoidal shapes to the left of the word "Cogna" in a black sans-serif font.The bolt logo features the word "bolt" in a white, bold, italicized sans-serif font on a dark blue rectangular background.

GitHub Spark



lovable



Theme 4: AI-Assisted App Development

- Narrows the scope to **app development**
- Usually narrows to particular **design theme** and **backend services**
- The user can **run the app** and **validate directly**
- Sometimes **little need to look at the code at all** (vibe coding)

Huge caveats

- **App coding** is replaced by **app description**
- This is a small but important slice of the software industry
- Good **UX design** is **subjective**, impossible to describe with words
- Beware security disasters!



Demo - GitHub Spark



General lessons for tool builders



Lesson: A small team can build a big thing

Introducing GitHub Copilot: your AI pair programmer

Today, we're launching a technical preview of GitHub Copilot, a new AI pair programmer that helps you write better code.

Technical preview

Your AI pair programmer

```
fetch_pic.js  push_to_git.py  JS_d3_scale.js  JS_fetch_stock.js  JS_material_ui.js

1 const fetchNASAPictureOfTheDay = () => {
2   return fetch('https://api.nasa.gov/planetary/apod?api_key=DEMO_KEY', {
3     method: 'GET',
4     headers: {
5       'Content-Type': 'application/json',
6     },
7   })
8   .then(response => response.json())
9   .then(json => {
10     return json;
11   });
12 }
```

 GitHub Copilot

Nat Friedman · @nat

June 29, 2021 | Updated February 23, 2022 | ⌚ 1 minutes

Share:   



Lesson: Build a prototype ASAP



Lesson: Deliver tangible results frequently

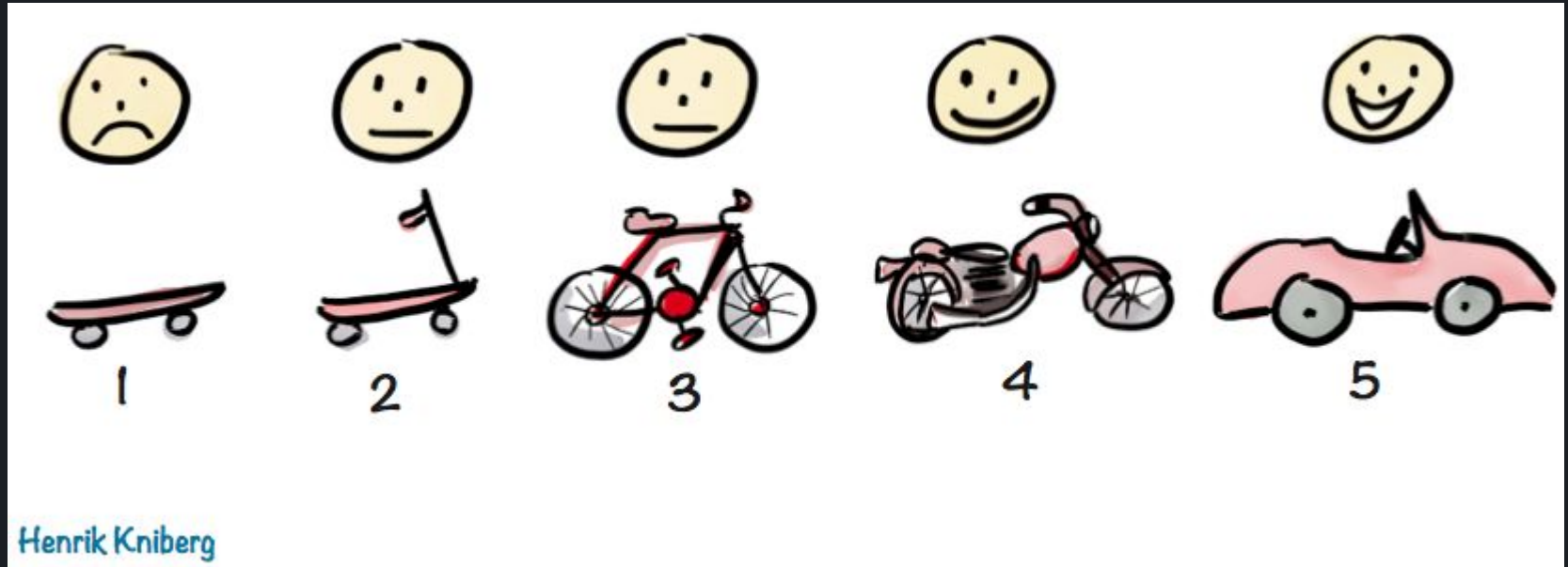


Image credit: <https://blog.crisp.se/2016/01/25/henrikkniberg/making-sense-of-mvp>



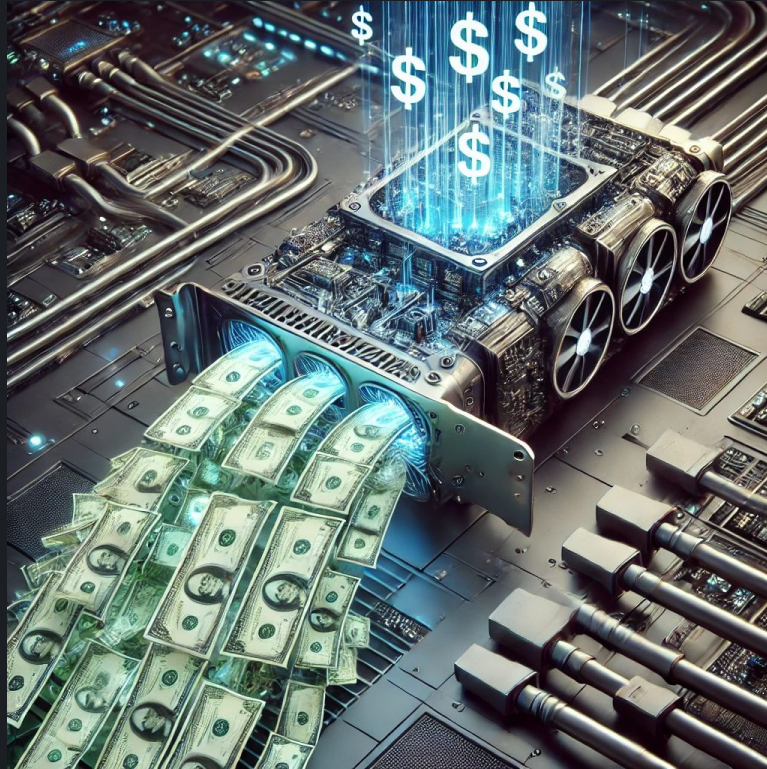
Lesson: UX is critical in AI developer tools



Lesson: Beware of long term market fit



Lesson: Don't over index on cost



Lesson: Dogfood intensively



Student Questions, My Answers



My Answers!

"What will coding with AI look like in the future?"

1. AI-powered **pinpoint tools** will be everywhere
2. AI-assisted **task orientation** will become normal
3. Some **technical coding** will be fully automated and become **technical description + exploration**
4. UX design is **subjective** and will be only partially automated
5. **Requirements, validation, architecture, maintainability, improvement, security** will be only partially automated



My Answers!

"Does AI mean that code will be obsolete soon?"

1. No! Code will largely remain the notation of record
2. **Technical description** will become more important
3. **Depth knowledge** will continue to be critical
4. **Requirements** and **validation** will begin to dominate



My Answers!

"How should I use AI tools in software development?"

1. Learn to be a great programmer **without AI**
2. **Embrace** many pinpoint tools
3. **Learn** task-oriented tools
4. **Play with** automation like Claude Code
5. **Play with** app-dev tools like Lovable and GitHub Spark



We live in incredible, disturbing times

- Coding is already magical, AI coding gives you wild powers
- These powers are not simple to use well
- AI is changing coding fast, and good developers embrace some of it

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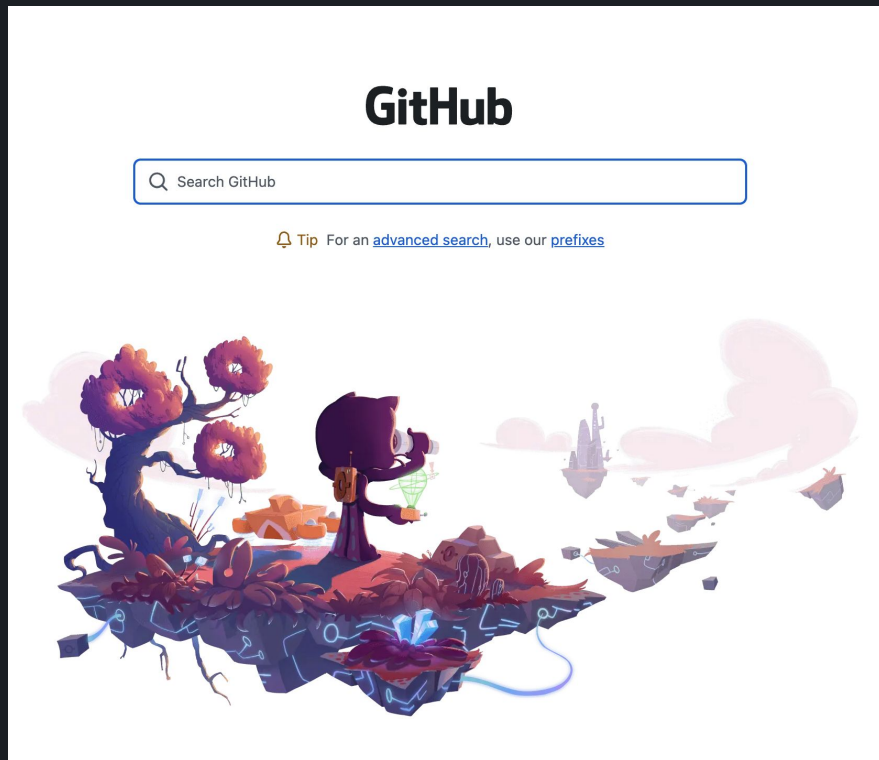
GitHub Next

Investigating the future of
software development

githubnext.com



Retrieval application 1: NL code search



Retrieval application #2: customized Copilot

GitHub Copilot for *Your* Codebase

When editing, GitHub Copilot only knows about the contents of your current file and possibly a few other open tabs, rendering it blind to important type definitions, patterns and greater connections in your codebase. We want to let GitHub Copilot see your entire repo when it comes up with its suggestions.

WHAT'S IT FOR?

Getting better, more relevant
Copilot suggestions

SHARE



STAGE

COMPLETED

WHO MADE IT?

 Eddie Aftandilian

 Johan Rosenkilde

 Amelia Wattenberger

 Vincent Hellendoorn



Retrieval application #3: Copilot for Docs

